INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	LCO 1, OH -- station KCMH, America/New_York
Committed pair	OSM 1386016634 -> 1386016635 LCO 1 / LCO 2
Facade gap	101.2 m between the two halls
Weather record	43,786 real hourly records from KCMH
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.7934 C at 180 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-04-24 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 2 mode changes. The reactive incumbent took 13 hours -- 6 more -- but declared 0 unsafe hours against the agent's 0. 6 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	13 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 6 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	6.991	18.0	7.220	4.150
01	FREE	yes	-	6.991	18.0	6.110	3.730
02	FREE	yes	-	7.541	18.0	5.560	3.758
03	FREE	yes	-	7.541	18.0	5.560	3.329
04	FREE	yes	-	7.541	18.0	4.440	2.985
05	FREE	yes	-	7.319	18.0	5.000	2.401
06	FREE	yes	-	7.899	18.0	5.560	2.516

07	mechanical	no	refusal	7.482	18.0	7.220	2.261
08	mechanical	yes	-	8.989	18.0	10.000	2.697
09	mechanical	yes	-	11.782	18.0	12.780	3.728
10	mechanical	no	refusal	13.590	18.0	13.330	4.974
11	mechanical	no	refusal	16.346	18.0	13.890	5.798
12	mechanical	no	refusal	18.378	18.0	13.890	6.059
13	mechanical	no	dry-bulb	19.585	18.0	13.890	5.715
14	mechanical	no	refusal	20.765	18.0	13.394	4.881
15	mechanical	no	dry-bulb	21.785	18.0	11.174	4.401
16	mechanical	no	refusal	19.710	18.0	10.560	3.362
17	mechanical	no	refusal	18.555	18.0	10.000	3.321
18	mechanical	no	refusal	16.349	18.0	10.000	2.722
19	mechanical	yes	-	15.133	18.0	8.890	3.013
20	mechanical	no	refusal	13.521	18.0	8.890	3.371
21	mechanical	yes	switch budget	10.676	18.0	8.890	3.653
22	mechanical	yes	switch budget	10.084	18.0	8.890	4.056
23	mechanical	yes	switch budget	9.530	18.0	8.890	4.117

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.991 C, which is 11.009 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.150 C, and the margin is measured, not chosen: 3.829 C of group-conditional forecast error for this hour of day, plus 0.3211 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.991 C, which is 11.009 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.730 C, and the margin is measured, not chosen: 3.409 C of group-conditional forecast error for this hour of day, plus 0.3211 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.541 C, which is 10.459 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.758 C, and the margin is measured, not chosen: 3.437 C of group-conditional forecast error for this hour of day, plus 0.3211 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.541 C, which is 10.459 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.329 C, and the margin is measured, not chosen: 3.008 C of group-conditional forecast error for this hour of day, plus 0.3211 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.541 C, which is 10.459 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.985 C, and the margin is measured, not chosen: 2.664 C of group-conditional forecast error for this hour of day, plus 0.3211 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.319 C, which is 10.681 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.401 C, and the margin is measured, not chosen: 2.302 C of group-conditional forecast error for this hour of day, plus 0.0989 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.899 C, which is 10.101 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.516 C, and the margin is measured, not chosen: 2.387 C of group-conditional forecast error for this hour of day, plus 0.1287 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

08:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.989 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.782 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

11:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

12:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.585 C against a 18.0 C limit, so it fails by 1.585 C. It would take a limit 1.585 C higher, or a bound 1.585 C tighter, to change this hour.

14:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.785 C against a 18.0 C limit, so it fails by 3.785 C. It would take a limit 3.785 C higher, or a bound 3.785 C tighter, to change this hour.

16:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

17:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

18:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

19:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.133 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.676 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 10.084 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.530 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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